WIRSCHING'S CONJECTURE 3, AND THE PERIODIC CORRECTION TO BERG–KRÜPPEL'S DENSITY ASYMPTOTIC

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Abstract. Wirsching's 2003 argument for positive predecessor density under the $3n + 1$ map reduces to three conjectures. Conjecture 3 concerns the near-zero asymptotics of an invariant density φ, compared against an explicit asymptotic φ_0 due to Berg and Krüppel. The exact log-Laplace transform of φ decomposes as a smooth part plus a periodic correction H. Berg and Krüppel gave this correction, for this eigenfunction, as an infinite product in 1998, five years before Wirsching's conjecture; they did not ask whether it is constant. We give H instead as a Fourier series with closed-form coefficients in Γ and ζ, and certify rigorously that it is not. Wirsching's own comparison class fixes a single phase of H; there, Conjecture 3 holds, with an explicit limit. Away from that class the phase sweeps a full period, so the stronger, unrestricted asymptotic $\varphi(t) \sim c\,\varphi_0(t)$ as $t \to 0^+$ fails. Wirsching needs less than the literal statement of Conjecture 3; we check directly that his actual requirement holds with a wide margin.

1. Introduction

Let T denote the $3n+1$ map on the positive integers, $T(n) = n/2$ for n even and $T(n) = (3n+1)/2$ for n odd. Wirsching's monograph [3] develops the predecessor-set machinery this paper's argument descends from; [1] takes it further, studying the density of predecessor sets under T through an averaging construction on the 3-adic integers: a family of operators S_l, built from path counts in the predecessor graph, converges in the strong operator topology to a limiting operator S_∞ whose essential part is a Markov chain on $[0, 1]$. That chain has a unique invariant density φ, and Wirsching's proof of positive predecessor density reduces to three conjectures about φ and related quantities, stated as three unbridged gaps in the argument.

The load-bearing one is Conjecture 3. It concerns the behaviour of φ as its argument tends to 0, compared against an explicit comparison function φ_0. Berg and Krüppel [2] had already produced φ_0 five years earlier, by Laplace transform and saddlepoint analysis, as the asymptotic solution of a truncated version of the functional equation φ satisfies. Their argument for the asymptotic comparability of φ and φ_0 was informal, and left open both the exact multiplicative constant and whether the ratio converges at all. Wirsching considers the strongest possible reading of that gap directly:

> "That replacement would be possible, if, e.g., $\varphi \sim c\,\varphi_0$ for some constant $c > 0$. It turns out that the following (slightly weaker) conjecture . . . suffices."

He then states, verbatim ([1], immediately after the quoted sentence; his own δ_5 is written δ throughout this paper, since no other δ-indexed class of his appears here):

Wirsching's Conjecture 3. *There are positive real constants c and δ such that*

$$\lim_{l\to\infty} \frac{\varphi(z_l)}{\varphi_0(z_l)} = c > 0 \qquad \textit{uniformly for sequences } (z_l) \in \widetilde{A}_\delta.$$

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Here $\widetilde{A}_\delta$ is Wirsching's class of sequences tending to 0 at the borderline rate $z_l \sim l \cdot 3^{-l}$ (his equations (1.5) and (3.2), reproduced ahead of Theorem 1's proof): a sequence (x_l) belongs to $\widetilde{A}_\delta$ exactly when $\lfloor 3^l x_l \rfloor = k_l$ for a sequence of integers (k_l) with $|l - k_l| \le \delta\sqrt{l}$ for every l. This is weaker than the unrestricted asymptotic he first considers. The constant cancels out of the one inequality his own proof actually needs, so a class-dependent limit is enough.

Neither statement had been settled. This paper settles both, in opposite directions.

Theorem 1 (Conjecture 3)**.** *Wirsching's Conjecture 3 (stated just above) is true. On his comparison class $\widetilde{A}_\delta$, the ratio $\varphi(z_l)/\varphi_0(z_l)$ converges uniformly to $e^{H(0)} = 0.534122\ldots$, where H is the periodic function of Theorem 4 below and the normalization of φ_0 is Berg and Krüppel's own, equation (9.6) of* [2].

Theorem 2 (Failure of the unrestricted asymptotic)**.** *The stronger asymptotic $\varphi(t) \sim \kappa\,\varphi_0(t)$ as $t \to 0^+$, for any single constant κ and with no restriction to a comparison class, is false. The ratio $\varphi(t)/\varphi_0(t)$ oscillates indefinitely as $t \to 0^+$, asymptotically* $\log 3$*-periodic in* $\log(1/t)$ *(periodic up to an $O(1/\tau)$-vanishing phase drift, Lemma 15) and with amplitude bounded below by a positive, certified constant.*

The two theorems have a single common source. Write $t = e^{-\tau}$ and let $H(w)$ be the $\log 3$-periodic function appearing in the exact decomposition of $\log\varphi$'s log-Laplace transform (Theorem 4); the saddlepoint bridge of Sections 4 and 5 shows $\log\varphi(t) - \log\varphi_0(t) = H(w_0(\tau)) + o(1)$ for a saddle location $w_0(\tau)$ that tracks τ up to a bounded shift. Wirsching's comparison class fixes $w_0(\tau)$'s phase modulo $\log 3$ at a single value, not τ's own phase, which the class does not control; every other class of comparison sequences sweeps $w_0(\tau)$'s phase through all values. Whether the ratio converges or oscillates depends on which sequences one compares along, and the two theorems above read the same object at two different scopes.

Corollary 3. *e^H is Berg and Krüppel's own periodic factor for the eigenfunction φ, from Proposition 9.3 of* [2]*, which they give as an infinite product; H itself is its logarithm. Its Fourier series, Proposition 6 below, is new.*

Section 2 recalls the functional equation φ satisfies and Berg and Krüppel's comparison asymptotic. Section 3 constructs H, proves Proposition 6 and Corollary 3, and certifies that H is not constant. Sections 4 and 5 transfer this from the log-Laplace transform to φ itself, through a uniform saddlepoint approximation and an envelope estimate. Section 6 identifies the resulting smooth part with Berg and Krüppel's saddlepoint expression and pins down every constant. Section 7 proves both theorems and checks Wirsching's actual requirement directly. Section 8 states what remains open.

2. The invariant density and its Laplace transform

Let $U_1, U_2, \ldots$ be independent, identically distributed uniform random variables on $[0,1]$, and set

$$X = \sum_{j\ge 1} \frac{2}{3^j} U_j \in [0,1].$$

The self-similar identity $X = \frac{2}{3}U_1 + \frac{1}{3}X'$, with X' an independent copy of X, shows that the density φ of X satisfies

$$\varphi(x) = \frac{3}{2}\int_{3x-2}^{3x} \varphi,$$

equivalently $\varphi'(x) = \frac{9}{2}\varphi(3x)$ on $(0, \frac{2}{3})$: this is Wirsching's invariant density (up to the reflection $x \mapsto 1-x$ fixing the convention), a translate of Rvachev's atomic function h_3 [4], and the dilation-3,

$\lambda = 2/3$ member of Berg and Krüppel's family of functional equations $\lambda\, g'(t) = 3\big(g(3t) - g(3t-2)\big)$. It is C^∞, supported on $[0,1]$, and constant, equal to $3/2$, on the middle third $[1/3, 2/3]$.

For $s > 0$ write

$$K(s) = \log \mathbb{E}\big[e^{-sX}\big] = \sum_{j\ge 1} g\left(\frac{2s}{3^j}\right), \qquad g(b) = \log \frac{1-e^{-b}}{b}.$$

The branch of g is the one analytic on $\{\operatorname{Re} w > 0\}$ and real on $(0,\infty)$; since $(1-e^{-w})/w$ is analytic and non-vanishing there (its zeros lie on $2\pi i\mathbb{Z} \setminus \{0\}$, and $w = 0$ is a removable singularity with value 1), and the half-plane is simply connected, g exists and is unique. Differentiating termwise, justified because $g(w) = -w/2 + O(w^2)$ as $w \to 0$ so the tail of the series is normally convergent together with every derivative,

$$t(s) := -K'(s), \qquad V(s) := s^2 K''(s) > 0,$$

the second an exact positivity, $K''(s)$ being the variance of X under its exponential tilt by e^{-sX}. Since $t'(s) = -K''(s) < 0$, t is strictly decreasing on $(0,\infty)$; at the endpoints, $t(0^+) = \mathbb{E}[X] = \sum_{j\ge1} 2{\cdot}3^{-j}{\cdot}\frac{1}{2} = \frac{1}{2}$, and $t(s) \to 0$ as $s \to \infty$ because the exponential tilt concentrates on $\operatorname{ess\,inf} X = 0$ (each factor e^{-sX} suppresses any mass away from 0 as $s \to \infty$, and $X \ge 0$ a.s.). So $t(s)$ decreases strictly from $1/2$ to 0, and studying $\varphi(t(s))$ as $s \to \infty$ is the same as studying $\varphi(t)$ as $t \to 0^+$.

Berg and Krüppel's comparison function solves the truncated equation $\lambda\, g'(t) = d\, g(dt)$ by Laplace transform and saddlepoint analysis. In their normalization (their equation (9.6), with $\alpha = \frac{1}{2} - \frac{\log 2}{\log 3}$, $\beta = \frac{1}{2\log 3}$, and γ, δ_{BK}, ε determined by α and β through their Proposition 9.1, printed explicitly in Section 6),

$$\varphi_0(t) = \frac{(2\beta)^\varepsilon}{\sqrt{2\pi}}\, t^\gamma\, (-\log t)^{\delta_{\mathrm{BK}}}\, \exp\left(-\beta \log^2 \frac{t}{-\log t}\right).$$

Write $\varphi_{0,\mathrm{bare}}$ for the same expression without the leading constant $(2\beta)^\varepsilon/\sqrt{2\pi}$: the two differ only by that constant.

3. The periodic correction

Set $c = \log 3$ and write $w = \log s$, $L(w) = K(e^w)$. Substituting into the defining series and splitting off the $j = 1$ term against the rest reproduces the recurrence

$$L(w+c) - L(w) = \log\big(1 - e^{-2e^w}\big) - w - \log 2. \tag{1}$$

Let

$$Q(w) = -\frac{w^2}{2c} + \left(\frac{1}{2} - \frac{\log 2}{c}\right) w,$$

chosen so that $Q(w+c) - Q(w) = -w - \log 2$, matching (1)'s non-periodic terms exactly. Write $a := \frac{1}{2} - \frac{\log 2}{c}$ for Q's linear coefficient, used again from Section 5 onward.

Theorem 4. *Define*

$$H(w) := L(w) - Q(w) + \sum_{k\ge0} \log\big(1 - e^{-2\cdot 3^k e^w}\big). \tag{2}$$

Then H is exactly c-periodic, $H(w+c) = H(w)$ for every $w \in \mathbb{R}$, and

$$L(w) = Q(w) + H(w) + \Delta(w), \qquad \Delta(w) = -\sum_{k\ge0} \log\big(1 - e^{-2\cdot3^k e^w}\big),$$

with $\Delta^{(j)}(w) = O_j\big(e^{jw - 2e^w}\big)$ for every $j \ge 0$: doubly exponentially small, together with all its derivatives, as $w \to +\infty$.

Proof. Write $d(w) = \log(1 - e^{-2e^w})$, so (1) reads $L(w+c) - L(w) = d(w) - w - \log 2$. Directly from the definition of Q (expand $(w+c)^2 = w^2 + 2wc + c^2$ and collect terms), $Q(w+c) - Q(w) = -w - \log 2$ as well, so the two non-periodic parts cancel exactly:

$$(L-Q)(w+c) - (L-Q)(w) = \big[d(w) - w - \log 2\big] - \big[-w - \log 2\big] = d(w).$$

The series in (2) telescopes this exactly: with $D(w) := (L-Q)(w)$,

$$D(w) + \sum_{k\geq 0} d(w+kc) = D(w+c) - d(w) + \sum_{k\geq 0} d(w+kc) = D(w+c) + \sum_{k\geq 0} d(w+(k+1)c),$$

so the right side of (2) is invariant under $w \mapsto w + c$, that is, H is c-periodic. Convergence of the series and its termwise differentiability, together with the stated bound on $\Delta^{(j)}$, follow from $d(w) = O(e^{-2e^w})$ and the corresponding bounds on its derivatives, since $2 \cdot 3^k e^w \geq 2e^w$ for $k \geq 0$ makes the series dominated, uniformly for w in any half-line bounded away from $-\infty$, by a geometric series in 3^{-k} inside the exponent. $\square$

The decomposition $L = Q + H + \Delta$, with Δ negligible as $w \to \infty$, is unique: any two periodic functions differing from $L - Q$ by an $o(1)$ term as $w \to \infty$ must be equal, being periodic functions that agree in the limit along every residue class of w modulo c.

Remark 5. Berg and Krüppel's own Proposition 9.3 (specialized to $a = 3$, $b = 3/2$ in their notation, exactly this eigenfunction) gives the equivalent identity $\exp(H(w)) = a^{\frac{1}{2}t^2 - \alpha t} \prod_{k\geq 0} \frac{1-e^{-a^{t-k}/b}}{a^{t-k}/b} \prod_{l\geq 1}(1 - e^{-a^{t+l}/b})$, $t = w/c$, as an infinite product, in their proof of that proposition. The Fourier series of Proposition 6 below is what lets us certify non-constancy (Proposition 8), which their product form does not make transparent.

Proposition 6. *H has the Fourier expansion $H(w) = \sum_{m\in\mathbb{Z}} \hat{H}(m) e^{i\omega_m w}$, $\omega_m = 2\pi m/c$, with*

$$\hat{H}(0) = \frac{\log 2}{2} - \frac{\log^2 2}{2c} - \frac{c}{12} - \frac{A}{c}, \qquad A = \frac{\pi^2}{12} - \frac{\gamma_E^2}{2} - \gamma_1,$$

$$\hat{H}(m) = -\frac{2^{i\omega_m}}{c}\,\Gamma(-i\omega_m)\,\zeta(1 - i\omega_m), \qquad m \neq 0,$$

where γ_E is the Euler–Mascheroni constant and γ_1 the first Stieltjes constant.

Sketch[1]*.* The Mellin transform of g on the strip $-1 < \operatorname{Re} z < 0$ is $g^*(z) = -\Gamma(z)\zeta(1+z)$, obtained by integrating by parts onto $g'(b) = 1/(e^b - 1) - 1/b$ and using the classical continuation $\int_0^\infty \big(\frac{1}{e^u - 1} - \frac{1}{u}\big) u^{z-1}\,du = \Gamma(z)\zeta(z)$. Since $K(s) = \sum_j g(2s/3^j)$, the Mellin transform of $s \mapsto K(s)$ picks up a factor $2^z \sum_j 3^{-jz} = 2^z/(3^z - 1)$ from this harmonic sum; shifting the resulting inversion contour past the triple pole at the origin and past each simple pole at $z = i\omega_m$ (where $3^z = 1$) reproduces $Q + \hat{H}(0)$ and the Fourier modes above, respectively, the factor $2^{i\omega_m}$ coming from that same harmonic-sum step. $\square$

Remark 7. This is an instance of the de Bruijn–Mahler phenomenon: a base-change harmonic sum, Mellin transformed, picks up simple poles on a vertical line, and their residues assemble into an exactly periodic correction with Fourier coefficients in Γ and ζ. De Bruijn found the first case of this in Mahler's partition asymptotic [5]; Erdős and Richmond name it on p. 448 of [6]; Kato and McLeod [7] and Derfel [8] established the log-squared decay rate of (2)'s companion asymptotic φ_0 as generic across this whole class of rescaled functional equations. Our contribution here is the closed form for the periodic correction itself, not the mechanism producing it.

[1] Proofs marked "Sketch" give the complete argument at the level of a referee report; every numerical constant they produce is independently certified in the accompanying repository (Section 9), and full line-by-line derivations are on record in the underlying project's notes, cited there.

The series for H' and H'' converge absolutely and uniformly, since $|\Gamma(-i\omega_m)| = \sqrt{\pi/(\omega_m \sinh \pi\omega_m)}$ decays exponentially in m while $\zeta(1-i\omega_m)$ grows only polynomially, and this yields explicit, certified bounds:

$$\sup_w |H'(w)| \le 0.0011977472315550332, \qquad \sup_w |H''(w)| \le 0.0068518962896650951. \tag{3}$$

Proposition 8 (Certified non-constancy)**.** *H is not constant. Rigorously,*

$$-0.000377190280943987 \; < \; H(0) - H(\log(3/2)) \; < \; -0.000377190280943985,$$

and consequently $\operatorname{osc}(H) := \sup H - \inf H \ge 3.7719 \times 10^{-4}$*; a full optimization gives the certified enclosure* $\operatorname{osc}(H) \in [4.18744947692 \times 10^{-4}, 4.18756644224 \times 10^{-4}]$.

Sketch. Write $g(b) = -b/2 + \log S(b/2)$, $S(x) = \sinh(x)/x = \sum_{n\ge 0} x^{2n}/(2n+1)!$ (this is g's defining formula rewritten around its removable singularity at 0, since $e^{-b/2}(e^{b/2} - e^{-b/2}) = 1 - e^{-b}$). Termwise domination of $6^n n! \le (2n+1)!$ gives $0 \le \log S(x) \le x^2/6$ for $x \ge 0$, hence, applied at $x = b/2$, an explicit, two-sided enclosure of the tail $\sum_{r>R} g(s/3^r)$ for any R and $s > 0$; combined with a finite head evaluated in interval or ball arithmetic (via (2) directly, or independently via ball arithmetic in Γ and ζ applied to Proposition 6), this certifies the stated interval to arbitrary precision. The full oscillation is certified the same way, via the Lipschitz bound in (3) and a fine grid search over one period. □

4. A uniform saddlepoint approximation

Define, for $s > 0$,

$$\varphi_{\mathrm{sp}}(t(s)) := \frac{\exp\bigl(K(s) + st(s)\bigr)}{\sqrt{2\pi K''(s)}}.$$

This section proves that $\varphi(t(s))/\varphi_{\mathrm{sp}}(t(s)) \to 1$ as $s \to \infty$, uniformly over every phase of s modulo $3^{\mathbb{Z}}$, which is what makes the two theorems of Section 7 depend on different scopes rather than on different mathematics.

For $n \ge 2$, define the derivative-normalized functions $\kappa_n(w) = (-1)^n w^n g^{(n)}(w)$ on $\{\operatorname{Re} w > 0\}$; $\kappa_n(b)$, $b > 0$, is b^n times the n-th cumulant of the random variable with density $b\, e^{-bv}/(1-e^{-b})$ on $[0, 1]$. Writing $x = w/2$,

$$\kappa_2(w) = 1 - \frac{x^2}{\sinh^2 x}, \qquad \kappa_3(w) = 2 - \frac{2x^3 \cosh x}{\sinh^3 x}, \qquad \kappa_4(w) = 6 + \frac{2x^4(1 - 3\coth^2 x)}{\sinh^2 x},$$

and from the Bernoulli expansion $g(w) + w/2 = \sum_{k\ge 1} B_{2k} w^{2k}/(2k(2k)!)$ on $|w| < 2\pi$,

$$\kappa_n(w) = (-1)^n \sum_{k \ge \lceil n/2 \rceil} \frac{B_{2k}\, w^{2k}}{2k\,(2k-n)!}, \qquad n \ge 2,$$

so that $\kappa_n(w) = O(w^2)$ as $w \to 0$ for every $n \ge 2$.

Write $s = \rho \cdot 3^N/2$ for the unique $N = \lfloor \log_3(2s) \rfloor \in \mathbb{Z}$ and $\rho \in [1, 3)$, and $b_j = 2s/3^j$. Then $\{b_j : j \ge 1\} = \{\rho \cdot 3^m : m \le N-1\}$; for $N \ge 1$ (the only range used from here on, corresponding to $s \ge 3/2$), exactly N of these, the smallest being $\rho \ge 1$, are ≥ 1, and the rest form a geometric tail below 1. This single structural fact is the source of every phase-independent constant below.

Lemma 9 (Boundedness off the real axis)**.** *For $w = b(1+iu)$, $b > 0$, $|u| \le 1$, with $e(r) = r^2/\sinh^2 r$ and $f(r) = r^3 \cosh r/\sinh^3 r$,*

$$|\kappa_2(w)| \le 1 + 2e(b/2), \qquad |\kappa_3(w)| \le 2 + 2 \cdot 2^{3/2} f(b/2).$$

Both bounds are finite over the whole sector $|\operatorname{Im} w| \le \operatorname{Re} w$; $M_2 := \sup|\kappa_2| \le 3$, $M_3 := \sup|\kappa_3| \le 2 + 2^{5/2} = 7.657$. In addition, for $|w| \le 2$, $|\kappa_2(w)| \le 0.114|w|^2$ and $|\kappa_3(w)| \le 0.0119|w|^4$.

Proof. $|\sinh z|^2 = \sinh^2(\operatorname{Re} z) + \sin^2(\operatorname{Im} z) \geq \sinh^2(\operatorname{Re} z)$ and $|\cosh z|^2 \leq \cosh^2(\operatorname{Re} z)$; the first excludes the poles of coth, since $\operatorname{Re}(w/2) = b/2 > 0$ never approaches $\pi i k$. This gives the sector bounds directly. Monotonicity of e is immediate from the increasing power series of $\sinh(r)/r$; monotonicity of f follows from $(\log f)'(r) = 3/r + \tanh r - 3\coth r < 0$ for $r > 0$ (Lazarević's inequality, $(\sinh r/r)^3 > \cosh r$, gives $f < 1$ but not this monotonicity by itself). The local bounds at $|w| \leq 2$ follow by applying the maximum-modulus principle to $\kappa_2(w)/w^2$ and $\kappa_3(w)/w^4$ (analytic on $|w| < 2\pi$ by the Bernoulli expansion): bound each on $|w| = 2$ by the triangle inequality, using $|B_{2k}| = 2(2k)!\,\zeta(2k)/(2\pi)^{2k}$ and $\zeta(2k) \leq \zeta(2)$, respectively $\zeta(4)$. □

Lemma 10 (Variance)**.** *For all $\rho \in [1, 3)$, $N \geq 1$, with $V := s^2K''(s) = \sum_j \kappa_2(b_j)$,*

$$N - A_V \leq V \leq N + 0.1283, \qquad A_V := \sum_{m\geq 0} e(3^m/2) = 1.4269413069\ldots.$$

Proof. Split $V = \sum_{m=0}^{N-1} \kappa_2(\rho 3^m) + \sum_{m<0} \kappa_2(\rho 3^m)$, using $\kappa_2(b) = 1 - e(b/2) \in (0, 1)$. For the lower bound, drop the second (non-negative) sum and use $e(\rho 3^m/2) \leq e(3^m/2)$ (e decreasing, $\rho \geq 1$) in the first, giving $\sum_{m=0}^{N-1} \kappa_2(\rho 3^m) \geq N - A_V$. For the upper bound, the first sum is $< N$ term by term, and the second sum, where $\rho 3^m < 1$, is bounded by Lemma 9's local estimate: $\sum_{m<0} \kappa_2(\rho 3^m) \leq 0.114\rho^2 \sum_{m<0} 9^m = 0.114\rho^2/8 \leq 0.1283$ for $\rho < 3$. □

Lemma 11 (Tail bound and existence of the density)**.** *If $b_0 \geq 1$ and $b_k = 3^k b_0$, then $\sum_{k\geq 0} \log\coth(b_k/2) \leq 3e^{-b_0}$. Consequently, with $\Lambda(y) := K(s(1 + iy)) - K(s) + isy\,t(s)$,*

$$|e^{\Lambda(y)}| \leq e^{3/e}(1 + y^2)^{-N/2} \leq 3.0152\,(1 + y^2)^{-N/2}$$

for every real y. In particular X has a continuous density, the tilted characteristic function is integrable for $N \geq 3$, and Fourier inversion at $x = t(s)$ gives

$$R(s) := \frac{\varphi(t(s))}{\varphi_{\mathrm{sp}}(t(s))} = \sqrt{\frac{V}{2\pi}} \int_{\mathbb{R}} e^{\Lambda(y)}\,dy, \tag{4}$$

a real quantity, since $\Lambda(-y) = \overline{\Lambda(y)}$.

Proof. $\log\coth(x/2) = 2\operatorname{artanh}(e^{-x}) \leq 2e^{-x}/(1 - e^{-2x})$; summing over the geometric orbit $b_k = 3^k b_0$ and bounding the resulting series by its first few terms gives the stated constant. X's density is continuous because the sum of the first two summands already has a continuous, compactly supported density and convolution preserves continuity. For Λ: $t(s) = -K'(s)$ cancels the linear term of the tilted transform exactly, and each remaining factor $e^{g(b_j(1+iy))-g(b_j)}$ has modulus at most 1 (a tilted characteristic function), with modulus at most $\coth(b_j/2)/\sqrt{1 + y^2}$ on the N indices where $b_j \geq 1$; the tail estimate applies to exactly that geometric set, with $b_0 = \rho \geq 1$. □

Lemma 12 (Central expansion)**.** *With $h(u) = g(b(1 + iu))$, Taylor's theorem in* integral *form (the Lagrange form is invalid for a complex-valued function of a real variable) gives*

$$h(y) = h(0) + h'(0)y + h''(0)\tfrac{y^2}{2} + \tfrac{1}{2}\int_0^y (y - u)^2 h'''(u)\,du.$$

Consequently, for $\theta \in (0, 1]$,

$$S := \sum_j \sup_{|u|\leq\theta} |h_j'''(u)| \leq 2N + 10.559,$$

and hence $|\Lambda(y) + Vy^2/2| \leq B|y|^3$ for $|y| \leq \theta$, with $B := S/6 \leq (2N + 10.559)/6$.

Proof. Write $h_j(u) := g(b_j(1+iu))$ and $w_j := b_j(1+iu)$. The chain rule and the definition of κ_3 give $h_j'''(u) = i\,\kappa_3(w_j)/(1+iu)^3$, so

$$|h_j'''(u)| = \frac{|\kappa_3(w_j)|}{(1+u^2)^{3/2}}.$$

Split the index set as in Section 4. For the N indices with $b_j = \rho 3^m \geq 1$, $0 \leq m \leq N-1$, drop the denominator and use Lemma 9's sector bound, legitimate since $|u| \leq \theta \leq 1$:

$$|h_j'''(u)| \leq |\kappa_3(w_j)| \leq 2 + 2^{5/2} f(b_j/2) \leq 2 + 2^{5/2} f(3^m/2),$$

the last step by monotonicity of f and $\rho \geq 1$. Summing over $m \geq 0$ bounds this block by $2N + 2^{5/2}\Sigma$, $\Sigma := \sum_{m\geq 0} f(3^m/2)$. The first three terms of Σ are $0.99614738\ldots$, $0.82241066\ldots$, $0.04500508\ldots$; for $r \geq 13$ one has $f(r) \leq 4.001\, r^3 e^{-2r}$, so the remaining terms sum to less than 2×10^{-8}, and $2^{5/2}\Sigma < 10.541906$.

For the remaining indices $b_j = \rho 3^{-n}$, $n \geq 1$, one has $b_j < 1$ and $|w_j| = b_j\sqrt{1+u^2} \leq b_j\sqrt{2} < 2$, so Lemma 9's local bound applies and, keeping the denominator this time,

$$|h_j'''(u)| \leq \frac{0.0119\,|w_j|^4}{(1+u^2)^{3/2}} = 0.0119\, b_j^4\,(1+u^2)^{1/2} \leq 0.0119\sqrt{2}\, b_j^4.$$

Summing, $\sum_{n\geq 1}(\rho 3^{-n})^4 = \rho^4/80 < 81/80$, so this block is below $0.0119\sqrt{2}\cdot 81/80 < 0.01704$. Together, $S < 2N + 10.541906 + 0.01704 < 2N + 10.559$.

For the remainder bound, $\sum_j h_j(y) = K(s(1+iy))$, $\sum_j h_j'(0) = isK'(s) = -is\,t(s)$ and $\sum_j h_j''(0) = -s^2K''(s) = -V$, so summing the integral-form Taylor expansion over j gives

$$\Lambda(y) + \frac{Vy^2}{2} = \sum_j \frac{1}{2}\int_0^y (y-u)^2 h_j'''(u)\,du,$$

whose modulus is at most $S|y|^3/6 = B|y|^3$ for $|y| \leq \theta$. □

Theorem 13 (Uniform saddlepoint asymptotic). *For $N \geq 19$, uniformly over $\rho \in [1,3)$, $|R(s) - 1| \leq E(N) := e_1(N) + e_2(N) + e_3(N)$, with $\theta_N := N^{-1/3}$, $V_{\mathrm{lo}} := N - A_V$, $V_{\mathrm{up}} := N + 0.1283$, $B := (2N + 10.559)/6$,*

$$e_1(N) := \frac{4\,e^{B\theta_N^3}\,B}{\sqrt{2\pi}\,V_{\mathrm{lo}}^{3/2}}, \qquad e_2(N) := \frac{2}{\sqrt{2\pi}}\cdot\frac{e^{-\theta_N^2 V_{\mathrm{lo}}/2}}{\theta_N\sqrt{V_{\mathrm{lo}}}},$$
$$e_3(N) := 2\,e^{3/e}\sqrt{\frac{V_{\mathrm{up}}}{2\pi}}\cdot\frac{(1+\theta_N^2)^{-(N-2)/2}}{\theta_N(N-2)}. \tag{5}$$

E is strictly decreasing, $E(19) < 1$, $E(N) \leq 4.18\,N^{-1/2}$ for all $N \geq 19$, and $\sqrt{N}\,E(N) \to \frac{4}{3}e^{1/3}(2\pi)^{-1/2} = 0.742358\ldots$. Consequently $\varphi(t) = \varphi_{\mathrm{sp}}(t)\,(1+o(1))$ as $t \to 0^+$, with no restriction on the sequence of t's.

Proof. Split (4) at $|y| = \theta_N$. On $|y| \leq \theta_N$, Lemma 12 bounds the cubic remainder $|\Lambda(y) + Vy^2/2|$ by $B\theta_N^3$, which stays bounded (not $o(1)$: this is exactly why e_1 carries the factor $e^{B\theta_N^3}$ rather than vanishing) as $N \to \infty$, since $B = O(N)$ and $\theta_N^3 = N^{-1}$. Write $C(y) := e^{\Lambda(y)} - e^{-Vy^2/2}$ and factor the Gaussian out before estimating, $e^{\Lambda(y)} = e^{-Vy^2/2}\,e^{z(y)}$ with $z(y) := \Lambda(y) + Vy^2/2$, so that $C(y) = e^{-Vy^2/2}\big(e^{z(y)} - 1\big)$. Since $|z(y)| \leq B|y|^3 \leq B\theta_N^3$ on $|y| \leq \theta_N$, the elementary $|e^z - 1| \leq |z|e^{|z|}$ gives

$$|C(y)| \;\leq\; e^{-Vy^2/2}\,B|y|^3\,e^{B\theta_N^3} \qquad (|y| \leq \theta_N),$$

the Gaussian factor retained. Integrating that inequality, $\int_{\mathbb{R}} e^{-Vy^2/2}|y|^3\,dy = 4/V^2$, so $\sqrt{V/2\pi}\int_{|y|\leq\theta_N}|C(y)|\,dy \leq 4Be^{B\theta_N^3}/(\sqrt{2\pi}\,V^{3/2})$, which is decreasing in V; using $V \geq V_{\mathrm{lo}}$ from Lemma 10 gives the central

Taylor-remainder contribution e_1. Since $\sqrt{V/2\pi}\int_{\mathbb{R}} e^{-Vy^2/2}\,dy = 1$, the same splitting leaves exactly two further terms. The truncated Gaussian tail $2\int_{\theta_N}^{\infty}\frac{1}{\sqrt{2\pi}}e^{-Vy^2/2}\,dy$, bounded via $\int_c^\infty e^{-v^2/2}\,dv \le e^{-c^2/2}/c$ at $c = \theta_N\sqrt{V_{\mathrm{lo}}}$, gives e_2. On $|y| > \theta_N$, Lemma 11's bound $|e^{\Lambda(y)}| \le e^{3/e}(1+y^2)^{-N/2}$ integrates, via $\int_a^\infty (1+y^2)^{-N/2}\,dy \le (1+a^2)^{-(N-2)/2}/(a(N-2))$ at $a = \theta_N$ (using $V \le V_{\mathrm{up}}$ for the leading $\sqrt{V/2\pi}$ factor in (4)), to give e_3; this needs $\theta_N \gg N^{-1/2}$, satisfied since $\frac{1}{3} < \frac{1}{2}$. Every constant entering E (via A_V, Lemma 10, and the $3.0152 \approx e^{3/e}$ of Lemma 11) is a supremum over $b > 0$ or a sum over one complete geometric orbit bounded using $\rho \ge 1$, hence independent of ρ. Evaluating (5) numerically gives the stated $N_0 = 19$, monotonicity, and limit; $E(18) = 1.00170\ldots$ and $E(19) = 0.95673\ldots$, and $\sqrt{19}\,E(19) = 4.1704\ldots$ with $\sqrt{N}E(N)$ decreasing. Since $t(s)$ ranges bijectively over $(0, 1/2)$, this is exactly the statement $\varphi(t) = \varphi_{\mathrm{sp}}(t)(1+o(1))$ as $t \to 0^+$. □

Remark 14. The rate is not sharp. Carrying the central expansion to fourth order, using that the odd cubic term integrates to zero against the Gaussian core, and using the limits $V_3 := \sum_j \kappa_3(b_j) = 2N + O(1)$, $V_4 := \sum_j \kappa_4(b_j) = 6N + O(1)$ (the third and fourth cumulant sums, by the same argument as Lemma 10) gives the Edgeworth correction $R(s) = 1 - \frac{1}{12N} + O(N^{-3/2})$, matching the classical Stirling correction for a Gamma$(N,1)$ law: after tilting, the $b_j \ge 1$ summands behave asymptotically like independent Exp(1) variables, so X under its tilt is asymptotically Gamma$(N,1)$-distributed, and ρ enters only through $O(1)$ shifts of the cumulants, at order N^{-2}. We do not need this sharper rate and do not prove the remainder term here; Theorem 13's qualitative $o(1)$ is all that Section 7 uses.

5. The envelope lemma

Theorem 13 compares φ to φ_{sp}, a quantity built from the full transform $K = Q + H + \Delta$. It remains to compare φ_{sp} to a smooth auxiliary system built from Q alone, so that the periodic part H can be extracted explicitly. Write $t = e^{-\tau}$, $g_\tau(w) = L(w) + e^{w-\tau}$, $g_{0,\tau}(w) = Q(w) + e^{w-\tau}$. The true and smooth systems each have their own analogue of $-K'$: write $B_{\mathrm{tr}}(w) := e^w t(e^w) = -L'(w)$ and $B_{\mathrm{sm}}(w) := -Q'(w) = w/c - a$.

Lemma 15 (True and smooth saddles)**.** *Let $\Phi(w) = w - \log B_{\mathrm{tr}}(w)$ and $\Phi_0(w) = w - \log B_{\mathrm{sm}}(w)$, and set $\tau_0 := 1 + ca + \log c = 0.9502067913\ldots$, so $\tau_0 > \log 2$.*

For $\tau > \log 2$, g_τ has a unique global minimizer $w^*(\tau)$, at which $g_\tau''(w^*) = V(r^*) > 0$, $r^* = e^{w^*}$; it is characterized by $\tau = \Phi(w^*)$, and Φ is a strictly increasing bijection $\mathbb{R} \to (\log 2, \infty)$. For $\tau \le \log 2$ there is no critical point at all. For $\tau > \tau_0$, $g_{0,\tau}$ has a unique local minimizer $w_0(\tau)$, characterized by $\tau = \Phi_0(w_0)$ together with $B_0 := B_{\mathrm{sm}}(w_0) = e^{w_0 - \tau} > 1/c$; Φ_0 restricted to $(ca+1, \infty)$ is a strictly increasing bijection onto (τ_0, ∞), and B_0 increases from $1/c$ to ∞ along it. For $\tau \le \tau_0$ there is no such minimizer.

Once $B_0 \ge 5$ (equivalently $\tau \ge 3.73978\ldots$),

$$|w^* - w_0| \le \frac{0.00652}{B_0}, \qquad 0 \le g_\tau(w_0) - g_\tau(w^*) \le \frac{5.77 \times 10^{-5}}{B_0}.$$

Proof. Substituting $s = e^w$ and $t = e^{-\tau}$ turns g_τ into $K(s) + st$, strictly convex in $s > 0$ because $K'' > 0$, with derivative $K'(s) + t$ vanishing exactly where $t = t(s)$. As $t(\cdot)$ decreases strictly from $1/2$ to 0, that happens for exactly one s when $t \in (0, 1/2)$, that is when $\tau > \log 2$, and for no s when $t \ge 1/2$. Since $w \mapsto e^w$ is a bijection $\mathbb{R} \to (0,\infty)$, $w^* = \log s$ is the unique global minimizer of g_τ, and Φ inverts $\tau \mapsto w^*$. Differentiating twice, $g_\tau''(w) = e^w\big(K'(e^w) + t\big) + e^{2w}K''(e^w)$, whose first term vanishes at w^*, leaving $g_\tau''(w^*) = V(r^*) > 0$.

For the smooth system, $g_{0,\tau}'(w) = e^{w-\tau} - B_{\mathrm{sm}}(w)$ and $g_{0,\tau}''(w) = e^{w-\tau} - 1/c$, so a critical point is a local minimizer exactly when $B_0 > 1/c$. Also $\Phi_0'(w) = 1 - 1/(c\,B_{\mathrm{sm}}(w))$, positive exactly where $B_{\mathrm{sm}}(w) > 1/c$, that is on $w > ca + 1$; there Φ_0 increases from $\Phi_0(ca+1) = ca + 1 + \log c = \tau_0$ to ∞. This gives the stated existence, uniqueness and range.

Now take $B_0 \geq 5$. The decomposition $L = Q + H + \Delta$ evaluates the true gradient at the smooth saddle exactly: $Q'(w_0) = -B_0 = -e^{w_0-\tau}$, so the leading terms cancel and

$$(6)\quad g_\tau'(w_0) = L'(w_0) + e^{w_0-\tau} = H'(w_0) + \Delta'(w_0), \qquad |g_\tau'(w_0)| \leq \varepsilon_1 := \sup|H'| + \sup_{[w_0-1,w_0+1]} |\Delta'|.$$

On $[w_0-1, w_0+1]$ write $g_\tau''(w) = e^{w-\tau} - 1/c + H''(w) + \Delta''(w)$. Here $e^{w-\tau} = B_0 e^{w-w_0}$ ranges over $[B_0/e,\, B_0 e]$, so g_τ'' admits no bound independent of τ; it is of exact order B_0,

$$(7)\qquad \frac{B_0}{e} - \eta_0 \;\leq\; g_\tau''(w) \;\leq\; B_0 e, \qquad \eta_0 := \frac{1}{c} + \sup|H''| + \sup_{[w_0-1,w_0+1]} |\Delta''|,$$

the upper bound because $-1/c + \sup|H''| + \sup|\Delta''| < 0$. Since $w_0 = c(B_0 + a)$, $B_0 \geq 5$ forces $w_0 \geq 5.3492$ and so $w \geq 4.3492$ on the interval, where Theorem 4's bound on $\Delta^{(j)}$ gives $|\Delta'|, |\Delta''| \leq 10^{-60}$. With (3) this makes $\varepsilon_1 \leq 0.0011978$ and $\eta_0 \leq 0.9170912$, whence $2e\eta_0 = 4.9859\ldots \leq 5 \leq B_0$, that is $\eta_0 \leq B_0/(2e)$, and (7) improves to $g_\tau'' \geq B_0/(2e) > 0$ throughout the interval.

Put $h_0 := 2e\varepsilon_1/B_0 \leq 2e \cdot 0.0011978/5 < 0.0014$. For $h \in (h_0, 1]$ the mean value theorem applied on $[w_0-1, w_0+1]$ gives $g_\tau'(w_0+h) \geq g_\tau'(w_0) + hB_0/(2e) \geq -\varepsilon_1 + hB_0/(2e) > 0$ and, symmetrically, $g_\tau'(w_0 - h) \leq \varepsilon_1 - hB_0/(2e) < 0$. As w^* is the only zero of g_τ', it lies in $(w_0 - h, w_0 + h)$ for every such h, so $|w^* - w_0| \leq h_0 = 2e\varepsilon_1/B_0 \leq 0.00652/B_0$.

Finally $g_\tau(w_0) - g_\tau(w^*) \geq 0$ because w^* is the global minimizer, and Taylor's theorem about w^*, where $g_\tau'(w^*) = 0$, bounds it by $\frac{1}{2}\big(\sup_{[w_0-1,w_0+1]} g_\tau''\big)(w_0 - w^*)^2 \leq \frac{1}{2}B_0 e\, h_0^2 = 2e^3\varepsilon_1^2/B_0 \leq 5.77 \times 10^{-5}/B_0$. ☐

Proposition 16 (Envelope estimate)**.** *With $S(\tau) := \log \varphi_{\mathrm{sp}}(t) = g_\tau(w^*) + w^* - \frac{1}{2}\log(2\pi V(r^*))$ and $P(\tau) := Q(w_0) + B_0 + w_0 - \frac{1}{2}\log\big(2\pi(B_0 - \frac{1}{c})\big)$,*

$$|S(\tau) - P(\tau) - H(w_0(\tau))| = O(1/\tau) \qquad (\tau \to \infty),$$

uniformly in the phase of τ modulo c.

Sketch. Expanding $S(\tau) - P(\tau) - H(w_0)$ exactly using $L = Q + H + \Delta$ and Lemma 15 produces four terms. The true envelope gap $g_{0,\tau}(w^*) - g_{0,\tau}(w_0)$ is $O(B_0\,(w^* - w_0)^2) = O(1/B_0)$, by Taylor about w_0 (where $g_{0,\tau}'$ vanishes) together with $g_{0,\tau}'' \leq B_0 e$ and $|w^* - w_0| = O(1/B_0)$, both from Lemma 15. The periodic-part difference $H(w^*) - H(w_0) = O(|w^* - w_0|) = O(1/B_0)$ using (3). The saddle-location shift $w^* - w_0$ itself, $O(1/B_0)$. Last, a normalization term comparing $V(r^*)$ to $B_0 - 1/c$: by the identity $V(r^*) = g_\tau''(w^*)$ and the expression for g_τ'' used in Lemma 15,

$$V(r^*) - \Big(B_0 - \frac{1}{c}\Big) = B_0\big(e^{w^*-w_0} - 1\big) + H''(w^*) + \Delta''(w^*) = O(1),$$

since $B_0|w^* - w_0| = O(1)$ and H'', Δ'' are bounded by (3) and Theorem 4; dividing by $B_0 - 1/c$ and taking $\frac{1}{2}\log(1 + O(1/B_0))$ gives $O(1/B_0)$ again. Since $\tau = w_0 - \log B_0$, $B_0/\tau \to 1/c$, and all four terms are $O(1/\tau)$ with constants free of the phase of τ. ☐

6. The Berg–Krüppel identity

Proposition 17. *Write $f(p) = \alpha \log p - \beta \log^2 p$, so that Berg and Krüppel's transform $G_0(p) = p^\alpha \exp(-\beta \log^2 p)$ of their equation (9.3) is $e^{f(p)}$ and their equation (9.5) reads $g_0(t) = (2\pi i)^{-1}\int_{\sigma-i\infty}^{\sigma+i\infty} \exp[pt + f(p)]\,dp$, $\sigma > 0$. Take α, β from their equation (9.4) at $a = 3$, $\lambda = 2/3$. Then $f(p) = Q(\log p)$ identically; the exact saddle of that integral, $t + f'(x) = 0$ with $t = e^{-\tau}$, is $x^\sharp = e^{w_0(\tau)}$; and*

$$P(\tau) \;=\; x^\sharp t + f(x^\sharp) - \tfrac{1}{2}\log\big(2\pi f''(x^\sharp)\big),$$

the saddlepoint expression invoked in the proof of their Proposition 9.1, evaluated at that exact saddle. In the variable $Y := cB_0$, with $r = -ca - \log c$, the saddle equation is $Y - \log Y = \tau + r$.

Proof. Their truncated equation (9.2) is $\lambda g'(t) = a\,g(at)$ for $a > 1$, $\lambda > 0$; Laplace transforming gives $\lambda p\,G(p) = G(p/a)$, whose general solution (9.3) is G_0 times an arbitrary 1-periodic function of $\log p/\log a$, with

$$\alpha = -\frac{1}{2} - \frac{\log\lambda}{\log a}, \qquad \beta = \frac{1}{2\log a}$$

their equation (9.4). Section 2 placed φ in that family at $a = 3$, $\lambda = 2/3$: their untruncated (9.1) reads $\frac{2}{3}\varphi'(t) = 3\big(\varphi(3t) - \varphi(3t-2)\big)$, which on $(0, \frac{2}{3})$ is $\varphi'(t) = \frac{9}{2}\varphi(3t)$. At those values

$$\alpha = -\frac{1}{2} - \frac{\log(2/3)}{\log 3} = \frac{1}{2} - \frac{\log 2}{\log 3} = a, \qquad \beta = \frac{1}{2\log 3} = \frac{1}{2c},$$

where the a on the right is this paper's linear coefficient of Q; their a is the dilation factor 3, and the collision of letters is unavoidable. Hence, writing $y := \log p$,

$$f(p) = \alpha y - \beta y^2 = ay - \frac{y^2}{2c} = Q(y),$$

the third of the three coincidences asserted below. The other two are the chain rule:

$$-p\,f'(p) = -Q'(y) = B_{\mathrm{sm}}(y) = 2\beta y - \alpha, \qquad p^2 f''(p) = Q''(y) - Q'(y) = B_{\mathrm{sm}}(y) - \frac{1}{c} = 2\beta y - 2\beta - \alpha.$$

Berg and Krüppel's own displayed formula for $f''(p)$, in the proof of their Proposition 9.1 (the unnumbered display right after "In view of"), reads $p^2 f''(p) = 2\beta\ln p + 2\beta - \alpha$, with $+2\beta$ where direct differentiation of f gives -2β: $f''(p) = \frac{1}{p^2}\big(2\beta\ln p - 2\beta - \alpha\big)$, since $f'(p) = \frac{1}{p}(\alpha - 2\beta\ln p)$ differentiates to $-\frac{2\beta}{p^2} - \frac{\alpha - 2\beta\ln p}{p^2}$. Their Proposition 9.1 is unaffected, the term being of lower order there than $2\beta\log p$, but the exact expression used here is not.

The saddle condition $t + f'(x) = 0$ at $x = e^y$ now reads $e^{-\tau} = B_{\mathrm{sm}}(y)e^{-y}$, that is $\tau = y - \log B_{\mathrm{sm}}(y) = \Phi_0(y)$, whose unique solution on the branch $B_{\mathrm{sm}} > 1/c$ is $y = w_0(\tau)$ by Lemma 15. So $x^\sharp = e^{w_0}$, and there $x^\sharp t = e^{w_0 - \tau} = B_0$, $f(x^\sharp) = Q(w_0)$, $(x^\sharp)^2 f''(x^\sharp) = B_0 - 1/c > 0$. Substituting,

$$x^\sharp t + f(x^\sharp) - \tfrac{1}{2}\log\big(2\pi f''(x^\sharp)\big) = B_0 + Q(w_0) - \tfrac{1}{2}\log\frac{2\pi(B_0 - 1/c)}{(x^\sharp)^2} = Q(w_0) + B_0 + w_0 - \tfrac{1}{2}\log\Big(2\pi\big(B_0 - \tfrac{1}{c}\big)\Big),$$

which is $P(\tau)$. Finally $w_0 = c(B_0 + a)$ and $\tau = w_0 - \log B_0$ give $\tau = cB_0 + ca - \log B_0 = Y - \log Y + ca + \log c = Y - \log Y - r$. $\square$

Series-reverting $Y - \log Y = \tau + r$ to second order, $Y = \tau + \log\tau + r + (\log\tau + r)/\tau + O(\tau^{-2}\log^2\tau)$, and substituting into P gives

$$P(\tau) = -\frac{(\tau + \log\tau)^2}{2c} + \Big(1 + \tfrac{1}{c} - \tfrac{r}{c}\Big)\tau + \Big(\tfrac{1}{2} - \tfrac{r}{c}\Big)\log\tau + C_P + O\bigg(\frac{\log^2\tau}{\tau}\bigg),$$

$$C_P = -\frac{r^2}{2c} + r + ca + \frac{ca^2}{2} - \frac{\log 2\pi}{2} + \frac{\log c}{2} = -0.9576743183133982760669122497855855\ldots.$$

Matching the τ and $\log\tau$ coefficients against Berg and Krüppel's γ, δ_{BK} (their equations following (9.6)) reproduces both exactly:

$$\gamma = -\Big(1 + \tfrac{1}{c} - \tfrac{r}{c}\Big) = -1.8649154949\ldots, \qquad \delta_{\mathrm{BK}} = \tfrac{1}{2} - \tfrac{r}{c} = 0.4546762683\ldots,$$

and, solving $C_P = \varepsilon\log(2\beta) - \frac{1}{2}\log(2\pi)$ for ε,

$$\varepsilon = a + \tfrac{1}{2} + \frac{\log c}{2c} = 0.4118732574\ldots, \qquad C_P = \varepsilon\log(2\beta) - \tfrac{1}{2}\log(2\pi) = \log\frac{(2\beta)^\varepsilon}{\sqrt{2\pi}},$$

so $P(\tau) - \log\varphi_{0,\mathrm{bare}}(e^{-\tau}) \to C_P$ and, using the prefactor-included normalization, $P(\tau) - \log\varphi_0(e^{-\tau}) \to 0$ with rate $O(\tau^{-1}\log^2\tau)$: this is precisely Berg and Krüppel's own Proposition 9.1 (asymptotic, by construction, not exact) applied to the identity of Proposition 17, not an independent asymptotic claim about P itself.

Proof of Corollary 3. By Proposition 16 and 17, $S(\tau) = \log \varphi_0(e^{-\tau}) + H(w_0(\tau)) + O(\tau^{-1} \log^2 \tau)$, i.e. $\varphi_{\rm sp} = \varphi_0 \cdot e^{H(w_0(\tau))}(1 + o(1))$: the periodic factor multiplying Berg and Krüppel's asymptotic solution, in equation (9.7) for this eigenfunction, is e^H, matching the product formula their Proposition 9.3 supplies for it, as noted after Theorem 4. Proposition 6's Fourier series for H is the new, closed form; Proposition 8's certificate is what neither their product formula nor their closing remark (that they only *expect* φ bounded and bounded away from zero relative to φ_0) settles. ☐

7. Proofs of the main theorems

Proof of Theorem 1. Wirsching's class $\widetilde{A}_\delta$ (his equations (1.5) and (3.2)) consists of sequences (x_l) with $\lfloor 3^l x_l \rfloor = k_l$ satisfying $|l - k_l| \le \delta\sqrt{l}$ for every l; hence $\lambda_l := 3^l x_l / l \to 1$ uniformly at rate $O(\delta/\sqrt{l})$ on the class. Writing $z_l = \lambda_l \, l \, 3^{-l}$, $\tau_l = -\log z_l = lc - \log l - \log \lambda_l$, the saddle location satisfies $w_0(\tau_l) = lc - \log \lambda_l + O(1/l)$ by Lemma 15's defining equation, so $w_0(\tau_l) \bmod c \to 0$ uniformly on the class, at rate $O(1/\sqrt{l})$. By Theorem 13, Proposition 16, and continuity of H (indeed Lipschitz, by (3)),

$$\log \varphi(z_l) - \log \varphi_0(z_l) = H(w_0(\tau_l)) + o(1) \to H(0)$$

uniformly on $\widetilde{A}_\delta$, i.e. $\varphi(z_l)/\varphi_0(z_l) \to e^{H(0)}$ uniformly. This is exactly Conjecture 3, with the constant given explicitly. ☐

Proof of Theorem 2. As $t = e^{-\tau}$ ranges over $(0, 1/2)$ with no restriction, $\tau \to \infty$, and Lemma 15 gives $w_0(\tau) \bmod c$ a full period (since Φ_0 restricted to $(ca+1, \infty)$ is a bijection onto (τ_0, ∞), and $\tau \mapsto \tau \bmod c$ composed with that inverse is onto one period once τ is large), so by the same combination of results used above, $\log \varphi(t) - \log \varphi_0(t) = H(w_0(\tau)) + o(1)$ takes values arbitrarily close to both $\sup H$ and $\inf H$ infinitely often. By Proposition 8, $\operatorname{osc}(H) > 0$ rigorously, so $\varphi(t)/\varphi_0(t)$ does not converge, and $\limsup / \liminf \ge e^{\operatorname{osc}(H)} \ge 1.0004188$. ☐

Proposition 18 (Wirsching's actual requirement)**.** *Write x_l^+ for the upper endpoint of Wirsching's own covering interval at level l (playing the same role as this paper's z_l). Wirsching's own use of Conjecture 3, via $\varphi'(x) = \frac{9}{2}\varphi(3x)$ on $(0, \frac{2}{3})$ and his own calculation, his equation (7.13), that $\lim_l (2/3^{l+1})\varphi_0'(x_l)/\varphi_0(x_l) = 2/3$ uniformly on the class, reduces to*

$$\limsup_l \Lambda_l < \tfrac{3}{2}, \qquad \Lambda_l := \frac{\varphi(3x_l^+)/\varphi_0(3x_l^+)}{\varphi(x_l^+)/\varphi_0(x_l^+)}.$$

This holds with a wide margin: $\Lambda_l \to 1$.

Proof. First, the reduction itself. Wirsching's actual requirement (his inequality (7.5)) is $\limsup_l (2/3^{l+1})\varphi'(x_l^+)/\varphi($ 1; by $\varphi' = \frac{9}{2}\varphi(3\cdot)$ this is $\limsup_l (9/3^{l+1})\varphi(3x_l^+)/\varphi(x_l^+) < 1$. By definition of Λ_l, $\varphi(3x_l^+)/\varphi(x_l^+) = \Lambda_l \cdot \varphi_0(3x_l^+)/\varphi_0(x_l^+)$, and Berg and Krüppel's φ_0 is an asymptotic solution of the same truncated equation (their equation (7.12) in Wirsching's notation: $\varphi_0'(t)/\varphi_0(3t) \to 9/2$ as $t \to 0$), so $\varphi_0(3x_l^+)/\varphi_0(x_l^+) = \frac{2}{9}\varphi_0'(x_l^+)/\varphi_0(x_l^+)\,(1 + o(1))$. Substituting and using his equation (7.13), $\lim_l (2/3^{l+1})\varphi_0'(x_l)/\varphi_0(x_l) = 2/3$ uniformly on the class,

$$\frac{9}{3^{l+1}} \cdot \frac{\varphi(3x_l^+)}{\varphi(x_l^+)} = \Lambda_l \cdot \frac{2}{3^{l+1}} \cdot \frac{\varphi_0'(x_l^+)}{\varphi_0(x_l^+)}\,(1 + o(1)) \to \frac{2}{3}\,\Lambda_l \cdot (1 + o(1)),$$

so the requirement $\limsup_l (9/3^{l+1})\varphi(3x_l^+)/\varphi(x_l^+) < 1$ is exactly $\limsup_l \Lambda_l < 3/2$.

For the value of Λ_l: the phase shift under $t \mapsto 3t$, i.e. $\tau \mapsto \tau - c$, is $w_0(\tau - c) - w_0(\tau) + c = O(1/\tau)$ by Lemma 15's defining equation, independently of the phase of τ. By Proposition 16, $\log \varphi(t) - \log \varphi_0(t) = H(w_0(\tau)) + O(1/\tau)$ uniformly in phase, so

$$\log \Lambda_l = H(w_0(\tau_l - c)) - H(w_0(\tau_l)) + O(1/\tau_l) = O(1/\tau_l) \to 0,$$

using that H is Lipschitz ((3)) and the phase shift above is $O(1/\tau_l)$; hence $\Lambda_l \to 1$, well inside $\limsup \Lambda_l < 3/2$. $\square$

8. Discussion

These results settle Conjecture 3 alone. Wirsching's positive-predecessor-density argument reduces to three conjectures; Conjectures 1 and 2, concerning respectively a pointwise-versus-average transfer for the Elka functions and a uniformity-near-a-singularity statement for the limiting operator S_∞, are untouched here, and the full theorem is not established by this paper.

Wirsching stated the weaker Conjecture 3 instead of the stronger $\varphi \sim \kappa\varphi_0$ he considers first. Theorem 2 shows why: the stronger statement fails by a mechanism with a certified, positive amplitude, not by some correctable technical looseness. The class on which the weaker statement holds is exactly the one phase, among the continuum swept by H, at which the ratio converges. That phase sits close to H's minimum: $H(0) - \min H = 5.25 \times 10^{-6}$ against a total oscillation of 4.19×10^{-4}. We do not know whether this is coincidental.

The rate is not established beyond $o(1)$. The sharper Edgeworth correction of Remark 14, and a correspondingly sharper form of the envelope and Berg–Krüppel comparisons, would give an explicit polynomial rate throughout; neither theorem needs it.

9. Code and data availability

Every certified or numerically-checked claim in this paper is backed by a runnable script in the accompanying repository, `https://github.com/faculdade/wirsching-conjecture3-proof`, organized by section with a README per script stating what it verifies and how to run it. This includes the interval-arithmetic certificates behind Proposition 8 and the bounds (3), the constant chain behind Theorem 13, and the identity of Proposition 17.

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